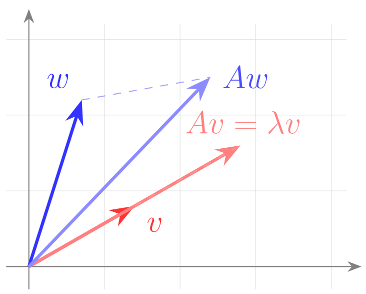
Exercise Sheet 11 – Solutions

Linear Algebra

What do the terms mean?

- **Eigenvalue λ and eigenvector** v: a number λ and a vector $v \neq 0$ with $Av = \lambda v$. That means: A only *stretches* the vector v (by the factor λ), without changing its direction.
- **Characteristic polynomial** $\chi_A(t) = \det(A - tI)$: the roots of this polynomial are exactly the eigenvalues. (Reason: $Av = \lambda v$ means $(A - \lambda I)v = 0$ with $v \neq 0$, so $A - \lambda I$ is *not* invertible, hence $\det(A - \lambda I) = 0$.)
- **Eigenspace** $E_\lambda = \ker(A - \lambda I)$: all eigenvectors for λ (plus the zero vector). You find it by solving the system $(A - \lambda I)v = 0$.
- **Algebraic multiplicity AM(λ)**: how often λ appears as a root of the char. polynomial (the exponent of $(t - \lambda)$).
- **Geometric multiplicity GM(λ)**: $\dim E_\lambda$ = number of basis vectors of the eigenspace = number of free variables when solving $(A - \lambda I)v = 0$.
- **It always holds that** $1 \leq \mathrm{GM}(\lambda) \leq \mathrm{AM}(\lambda)$. If all GM = AM, then A is *diagonalizable*.

[Graphical explanation – not part of the solution]



The basic idea: multiplying a vector by A usually rotates and *stretches it (blue: $w \to Aw$ points elsewhere). An* ***eigenvector*** *(red) is a special direction that A* only stretches *– Av lies on the same line as v. The stretch factor is the eigenvalue λ.*

Exercise 43

Problem. For $A = \begin{pmatrix} 4 & 1 & 1 \\ 2 & 5 & 2 \\ -1 & -1 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & 2 & 3 \\ 2 & 4 & 2 \\ -3 & -2 & -1 \end{pmatrix}$ determine the characteristic polynomial, the eigenvalues with algebraic and geometric multiplicity, and a basis of each eigenspace – in **two ways**.

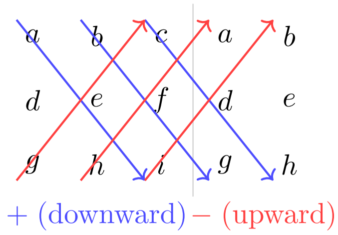
What needs to be done?

- **Method 1:** multiply out $\chi_A(t) = \det(A - tI)$ with the **Sarrus rule**, then factor with the **theorem on integer roots** (candidates = divisors of the constant term).
- **Method 2:** use **row/column operations** to factor out a term $(\lambda - t)$ early – this saves the multiplying-out.
- Then, per eigenvalue λ, solve the system $(A - \lambda I)v = 0 \Rightarrow$ basis of the eigenspace and GM.

Matrix A

Method 1: Sarrus + integer roots.

[Graphical explanation – not part of the solution]



Sarrus rule: write the first two columns again on the right. The three diagonals going down *(blue) count* $+$*, the three going* up *(red) count* $-$*. Formula:* $\det = aei + bfg + cdh - ceg - afh - bdi$.

Step 1: Write down $A - tI$. tI is the diagonal matrix with t all along the diagonal. So subtracting it changes *only the diagonal* of A:

$$A - tI = \begin{pmatrix} 4 & 1 & 1 \\ 2 & 5 & 2 \\ -1 & -1 & 2 \end{pmatrix} - \begin{pmatrix} t & 0 & 0 \\ 0 & t & 0 \\ 0 & 0 & t \end{pmatrix} = \begin{pmatrix} 4-t & 1 & 1 \\ 2 & 5-t & 2 \\ -1 & -1 & 2-t \end{pmatrix}.$$

Step 2: Apply Sarrus.

The rule in general. For an arbitrary 3×3 matrix with letters:

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \longrightarrow \det = aei + bfg + cdh \;-\; ceg - afh - bdi.$$

(The first three products = diagonals going down $(+)$, the last three = diagonals going up $(-)$, see the figure.)

Read off the letters from our matrix. From $A - tI = \left(\begin{smallmatrix} 4-t & 1 & 1 \\ 2 & 5-t & 2 \\ -1 & -1 & 2-t \end{smallmatrix}\right)$ we read entry by entry:

$$\begin{aligned} &a = 4-t, && b = 1, && c = 1, \\ &d = 2, && e = 5-t, && f = 2, \\ &g = -1, && h = -1, && i = 2-t. \end{aligned}$$

Compute the six products one by one:

$$\begin{aligned} aei &= (4-t)(5-t)(2-t), & ceg &= 1 \cdot (5-t) \cdot (-1) = -(5-t), \\ bfg &= 1 \cdot 2 \cdot (-1) = -2, & afh &= (4-t) \cdot 2 \cdot (-1) = -2(4-t), \\ cdh &= 1 \cdot 2 \cdot (-1) = -2, & bdi &= 1 \cdot 2 \cdot (2-t) = 2(2-t). \end{aligned}$$

Substitute into the formula (first three with $+$, last three with $-$):

$$\begin{aligned}\det(A-tI) &= aei+bfg+cdh-ceg-afh-bdi\\ &= (4-t)(5-t)(2-t)+(-2)+(-2)-\big(-(5-t)\big)-\big(-2(4-t)\big)-2(2-t)\\ &= (4-t)(5-t)(2-t)-2-2+(5-t)+2(4-t)-2(2-t).\end{aligned}$$

The bracket term multiplied out: $(4-t)(5-t)(2-t) = -t^3+11t^2-38t+40$. The remaining terms: $-2-2+5-t+8-2t-4+2t = 5-t$. Together:

$$\chi_A(t) = -t^3+11t^2-38t+40+5-t = -t^3+11t^2-39t+45.$$

[Explanation added] Quick check: constant term $\chi_A(0)=45=\det A$ ✓, and the t^2-coefficient $11=\operatorname{trace}(A)=4+5+2$ ✓.

Step 3: Factor. Factoring out -1: $\chi_A(t) = -(t^3-11t^2+39t-45)$. Integer roots are divisors of 45 – trying them shows: $t=3$ and $t=5$ give 0. The third root comes from the trace ($3+5+\lambda_3=11$): $\lambda_3=3$. So the roots are $\mathbf{3},\ \mathbf{3},\ \mathbf{5}$.

From roots to factored form. Each root λ becomes a factor $(t-\lambda)$ – a double root accordingly becomes a square:

$$t^3-11t^2+39t-45 = (t-3)\ \underbrace{(t-3)}_{\text{the 3 twice}}\ (t-5) = (t-3)^2(t-5).$$

We pull the -1 from before into the last factor, and since $(t-3)^2=(3-t)^2$ and $-(t-5)=(5-t)$, this becomes:

$$\boxed{\chi_A(t) = -(t-3)^2(t-5) = (3-t)^2(5-t).}$$

Exponent = algebraic multiplicity. The AM is simply how often the root occurs = the exponent of the factor: $\lambda=3$ is squared $\Rightarrow$ AM 2; $\lambda=5$ appears once $\Rightarrow$ AM 1.

Method 2: Factor out early (column trick).

[Graphical explanation – not part of the solution]

$$\begin{pmatrix}4-t & 1 & 1\\ 2 & 5-t & 2\\ -1 & -1 & 2-t\end{pmatrix} \xrightarrow{R_1+R_2+R_3} \begin{pmatrix}5-t & 5-t & 5-t\\ 2 & 5-t & 2\\ -1 & -1 & 2-t\end{pmatrix}$$

Observation: in A every column sums to 5 (e.g. $4+2-1=5$). So in $A-tI$ every column sums to $5-t$. If you add all rows together ($R_1\to R_1+R_2+R_3$), the top row becomes all $(5-t)$ – which you can pull out as a factor.

Step 1: $R_1 \to R_1+R_2+R_3$ (does not change the determinant), then factor out $(5-t)$:

$$\det(A-tI) = \det\begin{pmatrix}5-t & 5-t & 5-t\\ 2 & 5-t & 2\\ -1 & -1 & 2-t\end{pmatrix} = (5-t)\ \det\begin{pmatrix}1 & 1 & 1\\ 2 & 5-t & 2\\ -1 & -1 & 2-t\end{pmatrix}.$$

Step 2: columns $C_2\to C_2-C_1$ and $C_3\to C_3-C_1$ (makes the first row $1,0,0$):

$$= (5-t)\ \det\begin{pmatrix}1 & 0 & 0\\ 2 & 3-t & 0\\ -1 & 0 & 3-t\end{pmatrix} = (5-t)\,(3-t)(3-t) = (5-t)(3-t)^2.$$

[Explanation added] The last matrix is (lower) triangular $\Rightarrow$ determinant = product of the diagonal $1 \cdot (3-t) \cdot (3-t)$.

Same result as Method 1: $\chi_A(t) = (5-t)(3-t)^2$, eigenvalues $\lambda = 5$ (AM 1), $\lambda = 3$ (AM 2).

Eigenspaces.

$\lambda = 5$: solve $(A-5I)v = 0$. First $A - 5I$ (only the diagonal decreases by 5):

$$A - 5I = \begin{pmatrix} 4 & 1 & 1 \\ 2 & 5 & 2 \\ -1 & -1 & 2 \end{pmatrix} - \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix} = \begin{pmatrix} -1 & 1 & 1 \\ 2 & 0 & 2 \\ -1 & -1 & -3 \end{pmatrix}.$$

Then Gauss:

$$\begin{pmatrix} -1 & 1 & 1 \\ 2 & 0 & 2 \\ -1 & -1 & -3 \end{pmatrix} \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 + 2R_1} \begin{pmatrix} \boxed{-1} & 1 & 1 \\ 0 & \boxed{2} & 4 \\ 0 & -2 & -4 \end{pmatrix} \xrightarrow{R_3 \to R_3 + R_2} \begin{pmatrix} \boxed{-1} & 1 & 1 \\ 0 & \boxed{2} & 4 \\ 0 & 0 & 0 \end{pmatrix}.$$

Free variable z: from $2y + 4z = 0 \Rightarrow y = -2z$; from $-x + y + z = 0 \Rightarrow x = y + z = -z$. With $z = 1$:

$$E_5 = \operatorname{span}\{(-1,-2,1)\}, \qquad \mathrm{GM}(5) = 1.$$

$\lambda = 3$: First $A - 3I$:

$$A - 3I = \begin{pmatrix} 4 & 1 & 1 \\ 2 & 5 & 2 \\ -1 & -1 & 2 \end{pmatrix} - \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{pmatrix}.$$

Then Gauss:

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{pmatrix} \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 - 2R_1} \begin{pmatrix} \boxed{1} & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Only one equation $x + y + z = 0$, free variables y, z. With $(y,z) = (1,0)$ and $(0,1)$:

$$E_3 = \operatorname{span}\{(-1,1,0),\ (-1,0,1)\}, \qquad \mathrm{GM}(3) = 2.$$

$$\boxed{\begin{array}{c} \chi_A(t) = (5-t)(3-t)^2. \\ \lambda = 5\text{: AM 1, GM 1, basis } \{(-1,-2,1)\}. \quad \lambda = 3\text{: AM 2, GM 2, basis} \\ \{(-1,1,0),(-1,0,1)\}. \\ \text{All GM} = \text{AM} \Rightarrow A \text{ is diagonalizable.} \end{array}}$$

Matrix B

Method 1: Sarrus. First $B - tI$ (again only the diagonal):

$$B - tI = \begin{pmatrix} 5 & 2 & 3 \\ 2 & 4 & 2 \\ -3 & -2 & -1 \end{pmatrix} - \begin{pmatrix} t & 0 & 0 \\ 0 & t & 0 \\ 0 & 0 & t \end{pmatrix} = \begin{pmatrix} 5-t & 2 & 3 \\ 2 & 4-t & 2 \\ -3 & -2 & -1-t \end{pmatrix}.$$

Read off the letters (same formula $\det = aei + bfg + cdh - ceg - afh - bdi$ as for A):

$$\begin{array}{lll} a = 5-t, & b = 2, & c = 3, \\ d = 2, & e = 4-t, & f = 2, \\ g = -3, & h = -2, & i = -1-t. \end{array}$$

The six products:

$$\begin{aligned} aei &= (5-t)(4-t)(-1-t), & ceg &= 3\cdot(4-t)\cdot(-3) = -9(4-t), \\ bfg &= 2\cdot 2\cdot(-3) = -12, & afh &= (5-t)\cdot 2\cdot(-2) = -4(5-t), \\ cdh &= 3\cdot 2\cdot(-2) = -12, & bdi &= 2\cdot 2\cdot(-1-t) = 4(-1-t). \end{aligned}$$

Substitute into the formula:

$$\begin{aligned} \chi_B(t) &= aei + bfg + cdh - ceg - afh - bdi \\ &= (5-t)(4-t)(-1-t) + (-12) + (-12) - \big(-9(4-t)\big) - \big(-4(5-t)\big) - 4(-1-t) \\ &= (5-t)(4-t)(-1-t) - 12 - 12 + 9(4-t) + 4(5-t) - 4(-1-t) \\ &= (5-t)(4-t)(-1-t) + 36 - 9t = -t^3 + 8t^2 - 20t + 16. \end{aligned}$$

[Explanation added] Check: $\chi_B(0) = 16 = \det B$ ✓, t^2-coefficient $8 = \operatorname{trace}(B) = 5 + 4 - 1$ ✓.

Factor. $\chi_B(t) = -(t^3 - 8t^2 + 20t - 16)$. Test divisors of 16: $t = 2$ and $t = 4$ give 0. Third root from the trace ($2 + 4 + \lambda_3 = 8$): $\lambda_3 = 2$. Roots: **2**, **2**, **4**. As for A, each root becomes a factor, the double 2 a square:

$$\boxed{\chi_B(t) = -(t-2)^2(t-4) = (2-t)^2(4-t).}$$

Exponent = AM: $\lambda = 2$ squared $\Rightarrow$ AM 2; $\lambda = 4$ once $\Rightarrow$ AM 1.

Method 2: column trick. In B too every column sums to 4 ($5 + 2 - 3 = 4$, $2 + 4 - 2 = 4$, $3 + 2 - 1 = 4$). With $R_1 \to R_1 + R_2 + R_3$ and then $C_2 \to C_2 - C_1$, $C_3 \to C_3 - C_1$:

$$\det(B - tI) = (4-t)\det\begin{pmatrix} 1 & 0 & 0 \\ 2 & 2-t & 0 \\ -3 & 1 & 2-t \end{pmatrix} = (4-t)(2-t)^2.$$

Same result. Eigenvalues: $\lambda = 2$ (AM 2), $\lambda = 4$ (AM 1).

Eigenspaces.

$\lambda = 4$*:* First $B - 4I$:

$$B - 4I = \begin{pmatrix} 5 & 2 & 3 \\ 2 & 4 & 2 \\ -3 & -2 & -1 \end{pmatrix} - \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ -3 & -2 & -5 \end{pmatrix}.$$

Then Gauss:

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ -3 & -2 & -5 \end{pmatrix} \xrightarrow[R_3 \to R_3 + 3R_1]{R_2 \to R_2 - 2R_1} \begin{pmatrix} \boxed{1} & 2 & 3 \\ 0 & \boxed{-4} & -4 \\ 0 & 4 & 4 \end{pmatrix} \xrightarrow{R_3 \to R_3 + R_2} \begin{pmatrix} \boxed{1} & 2 & 3 \\ 0 & \boxed{-4} & -4 \\ 0 & 0 & 0 \end{pmatrix}.$$

$-4y - 4z = 0 \Rightarrow y = -z$; $x + 2y + 3z = 0 \Rightarrow x = -2y - 3z = -z$. With $z = 1$:

$$E_4 = \operatorname{span}\{(-1,-1,1)\}, \qquad \mathrm{GM}(4) = 1.$$

$\lambda = 2$: First $B - 2I$:

$$B - 2I = \begin{pmatrix} 5 & 2 & 3 \\ 2 & 4 & 2 \\ -3 & -2 & -1 \end{pmatrix} - \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 2 & 3 \\ 2 & 2 & 2 \\ -3 & -2 & -3 \end{pmatrix}.$$

Then Gauss:

$$\begin{pmatrix} 3 & 2 & 3 \\ 2 & 2 & 2 \\ -3 & -2 & -3 \end{pmatrix} \xrightarrow{R_2 \to \frac{1}{2}R_2} \begin{pmatrix} 3 & 2 & 3 \\ 1 & 1 & 1 \\ -3 & -2 & -3 \end{pmatrix} \xrightarrow[R_3 \to R_3 + 3R_2']{R_1 \to R_1 - 3R_2'} \begin{pmatrix} 0 & -1 & 0 \\ \boxed{1} & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

From the (-1) row: $-y = 0 \Rightarrow y = 0$; from $x + y + z = 0 \Rightarrow x = -z$. Only free variable z. With $z = 1$:

$$E_2 = \operatorname{span}\{(-1,0,1)\}, \qquad \mathrm{GM}(2) = 1.$$

$$\boxed{\begin{array}{c} \chi_B(t) = (2-t)^2(4-t). \\ \lambda = 2\text{: AM 2, GM 1, basis } \{(-1,0,1)\}. \quad \lambda = 4\text{: AM 1, GM 1, basis } \{(-1,-1,1)\}. \\ \text{Here } \mathrm{GM}(2) = 1 < \mathrm{AM}(2) = 2 \Rightarrow B \text{ is } \textit{not} \text{ diagonalizable.} \end{array}}$$

Exercise 44

Problem. Determine the eigenvalues (with AM, GM) and bases of the eigenspaces of $A = \begin{pmatrix} 1 & 0 & 5 \\ 0 & 6 & 0 \\ 1 & 0 & 5 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & 5 & 5 & 5 \\ 7 & 7 & 7 & 7 \\ 11 & 11 & 11 & 11 \\ -23 & -23 & -23 & -23 \end{pmatrix}$.

What needs to be done?

- **No characteristic polynomial!** Instead use properties: $\lambda = 0$ exactly when A is not invertible; sum of eigenvalues = trace; product = det; GM $\leq$ AM; read off obvious eigenvectors.

Matrix A

[Graphical explanation – not part of the solution]

$$\begin{array}{cl} & \text{row 1} \\ A = \begin{pmatrix} 1 & 0 & 5 \\ 0 & 6 & 0 \\ 1 & 0 & 5 \end{pmatrix} & Ae_2 = (0,6,0) = 6\,e_2 \Rightarrow \lambda = 6 \\ & = \text{row 3} \Rightarrow \det = 0 \Rightarrow \lambda = 0 \end{array}$$

Two things are immediate: (1) row 1 = row 3, so A is singular $\Rightarrow$ 0 is an eigenvalue. (2) The middle column is $(0,6,0)$, i.e. A sends $e_2 = (0,1,0)$ to $6\,e_2$ $\Rightarrow$ 6 is an eigenvalue with eigenvector e_2.

Step 1: $\lambda = 0$. Row 1 = row 3 $\Rightarrow$ $\det A = 0 \Rightarrow \lambda = 0$ is an eigenvalue. Eigenspace $= \ker A$:

$$Av = 0: \quad 6y = 0 \Rightarrow y = 0, \qquad x + 5z = 0 \Rightarrow x = -5z.$$

$$E_0 = \operatorname{span}\{(-5,0,1)\}, \qquad \mathrm{GM}(0) = 1.$$

Step 2: $\lambda = 6$. Eigenspace $= \ker(A - 6I)$. First

$$A - 6I = \begin{pmatrix} 1 & 0 & 5 \\ 0 & 6 & 0 \\ 1 & 0 & 5 \end{pmatrix} - \begin{pmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{pmatrix} = \begin{pmatrix} -5 & 0 & 5 \\ 0 & 0 & 0 \\ 1 & 0 & -1 \end{pmatrix}.$$

Both nontrivial rows say $x = z$ (y free). So

$$E_6 = \operatorname{span}\{(0,1,0),\ (1,0,1)\}, \qquad \mathrm{GM}(6) = 2.$$

Step 3: Add up the multiplicities. We have $\mathrm{GM}(6) = 2$, so $\mathrm{AM}(6) \geq 2$. Plus $\mathrm{AM}(0) \geq 1$. Since a 3×3 matrix has exactly 3 eigenvalues (with multiplicity), only $\mathrm{AM}(0) = 1$, $\mathrm{AM}(6) = 2$ remain. *[Explanation added]* Check with the trace: $0 \cdot 1 + 6 \cdot 2 = 12 = \operatorname{trace}(A) = 1 + 6 + 5$ ✓.

$$\boxed{\begin{array}{c} \lambda = 0\text{: AM 1, GM 1, basis } \{(-5,0,1)\}. \qquad \lambda = 6\text{: AM 2, GM 2, basis} \\ \{(0,1,0),(1,0,1)\}. \\ \text{All GM} = \text{AM} \Rightarrow A \text{ is diagonalizable.} \end{array}}$$

Matrix B

[Graphical explanation – not part of the solution]

$$B = \begin{pmatrix} 5 & 5 & 5 & 5 \\ 7 & 7 & 7 & 7 \\ 11 & 11 & 11 & 11 \\ -23 & -23 & -23 & -23 \end{pmatrix} \left.\vphantom{\begin{pmatrix} 5 \\ 7 \\ 11 \\ -23 \end{pmatrix}}\right\} \begin{array}{l} \text{every row} = c_i \cdot (1,1,1,1) \\ c = (5,7,11,-23) \\ \Rightarrow \text{rank } 1 \end{array}$$

*B has identical columns everywhere – every row is a multiple of $(1,1,1,1)$. Such a matrix has **rank** 1: all rows lie "on one line". Hence the kernel is huge (dimension $4 - 1 = 3$) and 0 is an eigenvalue.*

Step 1: $\lambda = 0$ with GM = 3. All columns of B are equal $\Rightarrow \operatorname{rank}(B) = 1 \Rightarrow \dim\ker(B) = 4 - 1 = 3$. So 0 is an eigenvalue with $\mathrm{GM}(0) = 3$. The kernel: $Bv = 0$ means (each row) $5(a+b+c+d) = 0$, i.e. $a+b+c+d = 0$:

$$E_0 = \operatorname{span}\{(-1,1,0,0),\ (-1,0,1,0),\ (-1,0,0,1)\}, \qquad \mathrm{GM}(0) = 3.$$

Step 2: There is no further eigenvalue. The sum of all eigenvalues $= \operatorname{trace}(B) = 5 + 7 + 11 - 23 = 0$. So far we have $\lambda = 0$ with $\mathrm{AM}(0) \geq \mathrm{GM}(0) = 3$. If the fourth eigenvalue μ were $\neq 0$, the sum would be $\mu \neq 0$ – contradiction. So $\mu = 0$ and

$$\mathrm{AM}(0) = 4.$$

[Explanation added] Put differently: a rank-1 matrix $c\,r^\top$ has only the eigenvalue $r^\top c = \text{trace}$ (once) and otherwise 0. Here trace $= 0$, so *all* eigenvalues are 0 (the matrix is nilpotent, $B^2 = 0$).

$\lambda = 0$: AM 4, GM 3, basis $\{(-1, 1, 0, 0), (-1, 0, 1, 0), (-1, 0, 0, 1)\}$.
$\mathrm{GM}(0) = 3 < \mathrm{AM}(0) = 4 \Rightarrow B$ is *not* diagonalizable.

Exercise 45

Problem. $A, B \in \mathbb{F}^{n\times n}$. Prove the following claims.

What needs to be done?

- In each claim we start with "λ is an eigenvalue of A" = "there is $v \neq 0$ with $Av = \lambda v$" and check that the same v is also an eigenvector of the new matrix.

(a) If λ is an eigenvalue of A and $c \in \mathbb{F}$, then $\lambda + c$ is an eigenvalue of $A + cI$.
Proof. Let $v \neq 0$ with $Av = \lambda v$. Then

$$(A + cI)v = Av + c\,v = \lambda v + cv = (\lambda + c)\,v.$$

Since $v \neq 0$, $\lambda + c$ is an eigenvalue of $A + cI$ (with the same eigenvector v). □

(b) If λ is an eigenvalue of A, then λ^2 is an eigenvalue of A^2.
Proof. Let $v \neq 0$ with $Av = \lambda v$. Then

$$A^2 v = A(Av) = A(\lambda v) = \lambda(Av) = \lambda(\lambda v) = \lambda^2 v. \qquad \square$$

(c) If A is invertible and λ is an eigenvalue of A, then λ^{-1} is an eigenvalue of A^{-1}.

Proof. Since A is invertible, 0 is not an eigenvalue (otherwise there would be $v \neq 0$ with $Av = 0$, so A would not be injective). Hence $\lambda \neq 0$. From $Av = \lambda v$, applying A^{-1}:

$$v = A^{-1}(\lambda v) = \lambda\, A^{-1} v \implies A^{-1} v = \tfrac{1}{\lambda}\, v.$$

Since $v \neq 0$, λ^{-1} is an eigenvalue of A^{-1}. □

(d) If v is an eigenvector of both A and B, then v is also an eigenvector of AB.

Proof. Let $v \neq 0$ with $Av = \lambda v$ and $Bv = \mu v$. Then

$$(AB)v = A(Bv) = A(\mu v) = \mu(Av) = \mu(\lambda v) = (\lambda\mu)\, v.$$

Since $v \neq 0$, v is an eigenvector of AB for the eigenvalue $\lambda\mu$. □

Overview: Results 43 & 44

Matrix	Eigenvalue (λ)	AM, GM	Basis of the eigenspace
43 A	3	2, 2	$(-1,1,0)$, $(-1,0,1)$
	5	1, 1	$(-1,-2,1)$
43 B	2	2, 1	$(-1,0,1)$
	4	1, 1	$(-1,-1,1)$
44 A	0	1, 1	$(-5,0,1)$
	6	2, 2	$(0,1,0)$, $(1,0,1)$
44 B	0	4, 3	$(-1,1,0,0)$, $(-1,0,1,0)$, $(-1,0,0,1)$

Diagonalizable are 43 A and 44 A (all GM = AM). *Not* diagonalizable are 43 B and 44 B (one GM $<$ AM each).